

## GPU (Unified Memory)

MIG Instance A (12 SMs)

MIG Instance B (6 SMs)

*reads and writes in place*

## Serving Processes (One Address Space Each)

Publisher

Weights KV Cache  
model prefix tail

①

Agent 1

Weights KV Cache  
model prefix tail

Agent 2

Weights KV Cache  
model prefix tail

②

③

Agent 3

Weights KV Cache  
model prefix tail

Host Page Tables: one per process, walked by the GPU through the SMMU

*4 mappings, 1 copy*

*4 mappings, 1 copy*

*1 mapping each*

Model File

*one copy, 2 MiB pages*

Prefix Cache File (tmpfs)

*one copy, immutable rows*

*names the rows*

State File: metadata only



Private Tails

*anonymous memory,  
grows with the rows in use*

① publish

② attach

③ grow the tail

— read-only mapping, shared pages

— writable mapping, pages of one process

Host Memory (One DRAM)